SE375 SYSTEM PROGRAMMING SPRING 2025-2026
Laboratory Assignment 3

March 13, 2026

1. Introduction

Assume that each district (Konak, Gaziemir etc) of İzmir reports their data on public transport usage and the overall data set is collected for analysis. The data set from each district contains the daily number of bus passengers and number of bus trips. An example view of a data set from one district (say Konak) is as following;

Date	Day of the Week	Number of Trips	Number of Passengers
01.02.2023	Wednesday	34	1242
02.02.2023	Thursday	51	4589
03.02.2023	Friday	30	2568
04.02.2023	Saturday	48	3874
05.02.2023	Sunday	36	2514
06.02.2023	Monday	44	6257
07.02.2023	Tuesday	27	2674

2. Task

In this lab assignment, you are going to develop a Java Application that will calculate statistics on given data. Your application will read each data file, compute and store the values and report to the user.

a. Specifications

- Each data file contains the daily number of bus trips and number of bus passengers in a CSV (comma-separated values) format for each district. The first line of the file has the headers in the order of:

  Date, *Day of the Week*, *Number of Trips* and *Number of Passengers*

  and the remaining lines include data for each date of the month.
- `DD.MM.YYYY` is the date format.
- Day of the week is one of the below;

Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday

- The data type of *Number of Trips* and *Number of Passengers* is integer. It is guaranteed that the data files do not have missing values.

- Your application should use a single `HashMap` to store the statistics where keys are the dates (e.g., 01.02.2023), and the values are:
    - The total number of passengers on that date over all given districts,
    - The average number of passengers on that date over all given districts.
- Your application should create a thread for each data file and give both the file name and the `HashMap` as an argument. This structure should be shared between all threads.
- Each thread should **read** the given data file.
- Each thread should **update** the `HashMap` while processing the given data file.
- Once threads are finished, your application should **display** the data in the `HashMap`. The main application should not update the `HashMap` at any point.

b. Example Runs

The result should look like the following:

```
Parsing Karabaglar.csv...
Parsing Konak.csv...
Parsing Gaziemir.csv...
Parsing Bornova.csv...
Parsing Urla.csv...
Parsing Karsiyaka.csv...
Parsing Balcova.csv...
Parsing Seferihisar.csv...
Threads are complete. Here are the statistics for the number of passengers:

Date                Total Passengers     Avg Passengers
----                ----------------     --------------
15.03.2023          19839                2479.88
14.03.2023          19766                2470.75
13.03.2023          19756                2469.5
16.03.2023          18509                2313.63
30.03.2023          19251                2406.38
17.03.2023          20629                2578.63
12.03.2023          18143                2267.88
18.03.2023          20962                2620.25
10.03.2023          18763                2345.38
04.03.2023          19975                2496.88
11.03.2023          20287                2535.88
25.03.2023          19232                2404.0
19.03.2023          20136                2517.0
24.03.2023          19095                2386.88
05.03.2023          19018                2377.25
06.03.2023          19146                2393.25
23.03.2023          18032                2254.0
08.03.2023          20322                2540.25
22.03.2023          16708                2088.5
07.03.2023          20231                2528.88
21.03.2023          19823                2477.88
09.03.2023          18781                2347.63
20.03.2023          20955                2619.38
26.03.2023          21178                2647.25
03.03.2023          18252                2281.5
27.03.2023          21141                2642.63
02.03.2023          18362                2295.25
31.03.2023          20061                2507.63
28.03.2023          18289                2286.13
01.03.2023          21252                2656.5
29.03.2023          19012                2376.5
```

These are the correct results; you may compare your results with these numbers to check whether they are correct.

The results might be displayed in a different order since threads may have a different running order at each run of the program, and since the `HashMap` is not an ordered structure.

3. Regulations

- You should use Java.
- Late submission is **not** allowed.
- This is an **individual** assignment. Plagiarism, copying, cheating, outsourcing the exam to another person or organization for with or without pay are considered as actions of academic dishonesty. Failure to maintain academic honesty may result in disciplinary action according to the Izmir University of Economics' disciplinary bylaw for students of institutions of higher education (https://www.ieu.edu.tr/en/bylaws/type/read/id/13).
- Submissions should be done to Blackboard. Submissions via e-mail will not be accepted.
- You should make sure to obey the specifications.

4. Submission

- You should submit java files to Blackboard.
- By submitting you pledge the following; *"On my honor, as an Izmir University of Economics student, I affirm that I have not given or received any unauthorized help on this exam, and that this work is my own."*